Predicting Singapore HDB Monthly Rent

CS5228 Project Progress Report – Group 44

Han Linxuan	A0350911M
Sun Shenghe	A0354849L
Zhang Yuhong	A0356795H
Zhang Zhexuan	A0355290A

School of Computing
National University of Singapore

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Problem and project goals

Prediction task

Estimate the monthly rent of an HDB flat from its approval month, location, flat characteristics and permitted auxiliary data.

- Metric: root mean squared error (RMSE)
- Output: one rent prediction for each of 50,000 test rows
- Main challenge: future rents must be predicted from historical transactions

Source: CS5228 project page and Kaggle competition description

Core research question

Can historical comparable rents improve future HDB rental forecasts as market conditions change?

We also examine location effects, temporal stability and the sources of large prediction errors.

Data and prediction horizon

Core data

- 150,000 labelled rows
- Jan 2021 to Mar 2025
- 50,000 test rows
- Mar 2025 to Jul 2026
- Target range: SGD 300 to 7,600

Auxiliary data

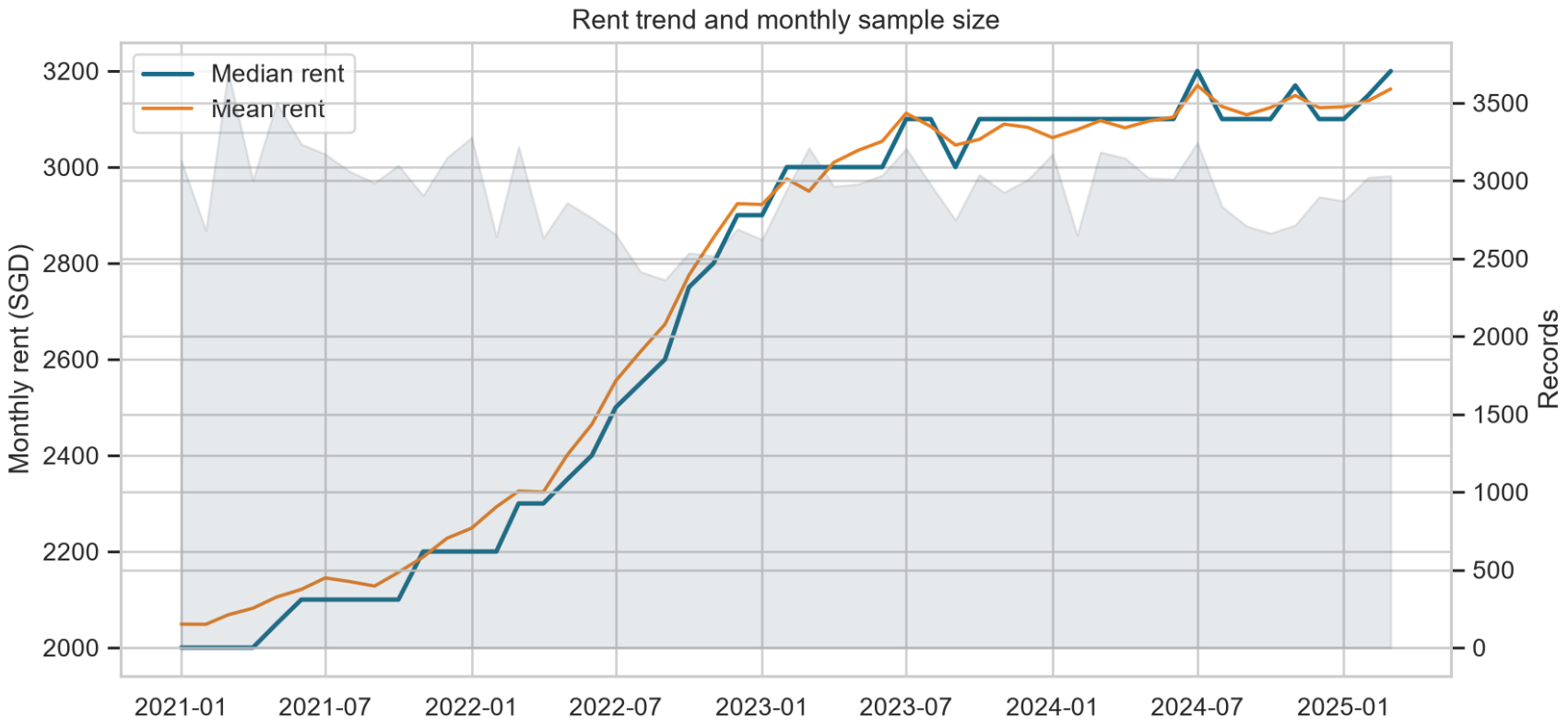
- HDB block coordinates and attributes
- MRT stations, malls and primary schools
- COE and stock-price indicators (tested, not retained)

Consequence

The test dates cover a 17-month span ending in Jul 2026. Time extrapolation matters as much as model fit.

Source: Official course dataset; local data audit

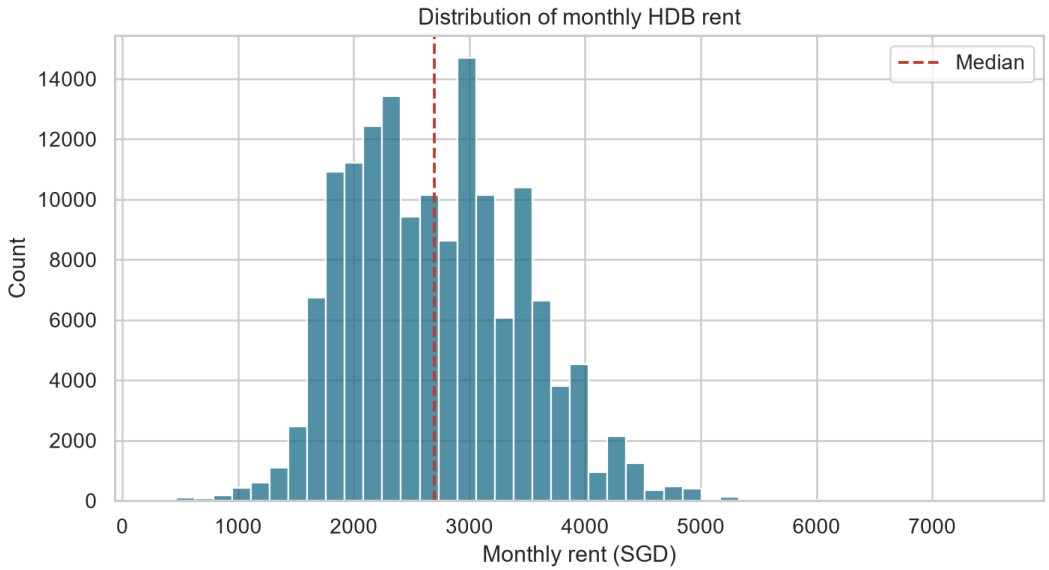
Rental levels changed substantially over time



Median rent rose from about SGD 2,000 in early 2021 to around SGD 3,200 by early 2025. A random split would mix different market regimes and overstate future performance.

Source: Reproducible EDA on the 150,000 training rows

Rent varies across locations and contains difficult tails



Overall median: SGD 2,700

Town medians

Town	SGD
Central	3,000
Bukit Timah	3,000
Bukit Merah	2,900
Ang Mo Kio	2,500
Bedok	2,500
Woodlands	2,500

Town medians differ, but this comparison also reflects date and flat-mix differences.

The distribution is multi-modal and includes unusually low rents, so we also analyse errors by rent band.

Source: Reproducible EDA summaries on the 150,000 training rows

Preprocessing and feature design

Data preparation

- Canonicalise text and flat-type formats
- Remove constant inputs: furnished and fee
- Audit 425 exact duplicate rows without automatic deletion
- Preserve raw files; validate schemas, hashes and output IDs

Feature groups

- Time, flat size, model and lease age
- Block, town, planning area and region
- Distance and counts for MRT, malls and schools
- Past-only comparable rents and local market momentum

Leakage control

Historical rent features use only months strictly earlier than each predicted row.

Source: Project feature pipeline and automated data-contract tests

Chronological evaluation design

Primary comparison

Expanding training data followed by future validation windows. The latest split trains on 141,076 rows through Dec 2024 and validates on 8,924 rows from Jan–Mar 2025.

- Matched rows and preprocessing within each comparison
- Later screens fix seeds and 1,600-tree budgets
- Rolling three-month folds plus older-year and sixteen-month stress windows
- RMSE for selection; MAE and signed bias for diagnosis

Source: Project validation code and Kaggle evaluation description

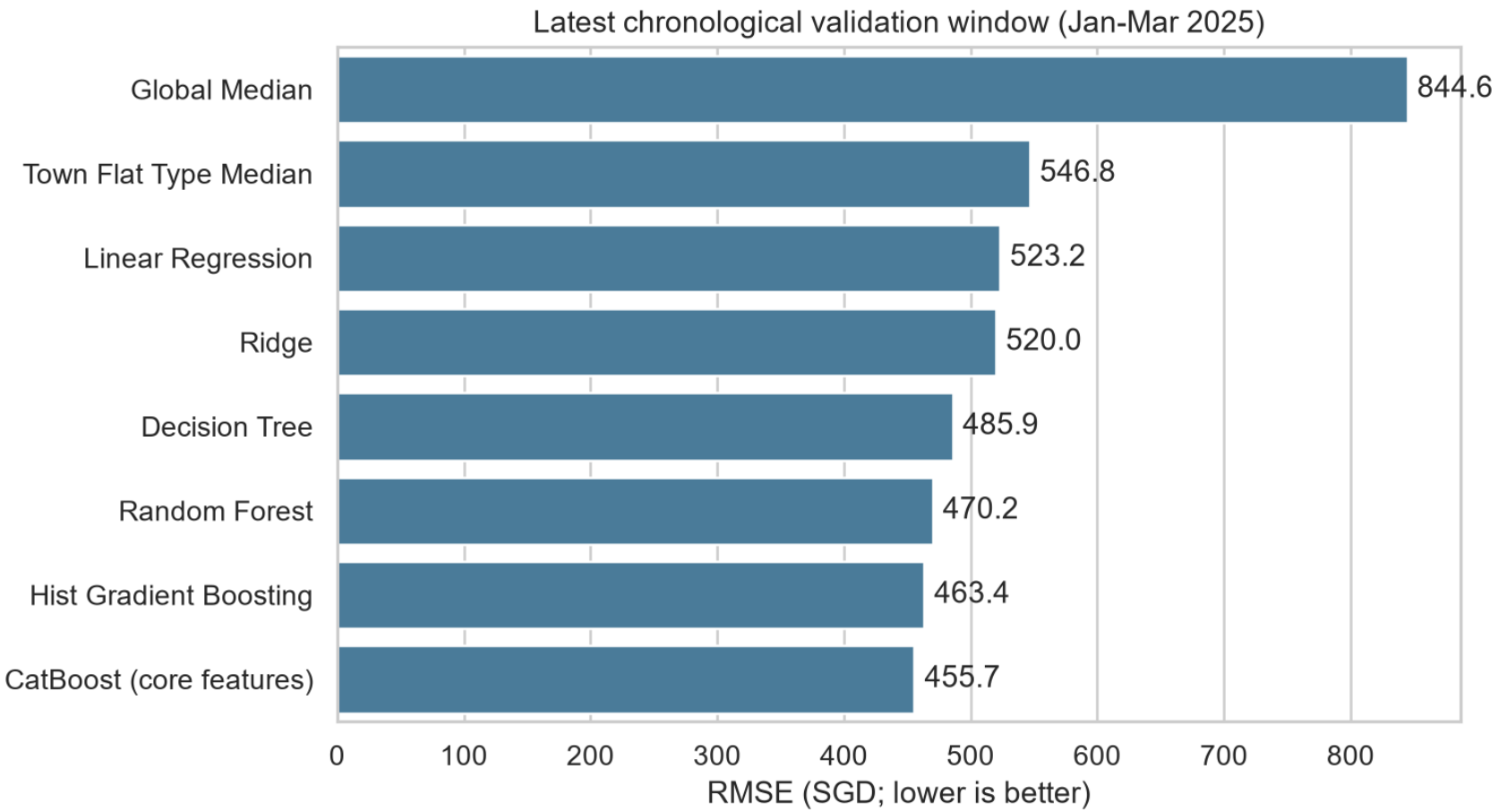
Why this matters

The public leaderboard covers about half of the hidden test set. Local stability reduces dependence on one public split.

Interpretation

Historical features never use validation labels. Overlapping stress windows test robustness but are not independent test sets.

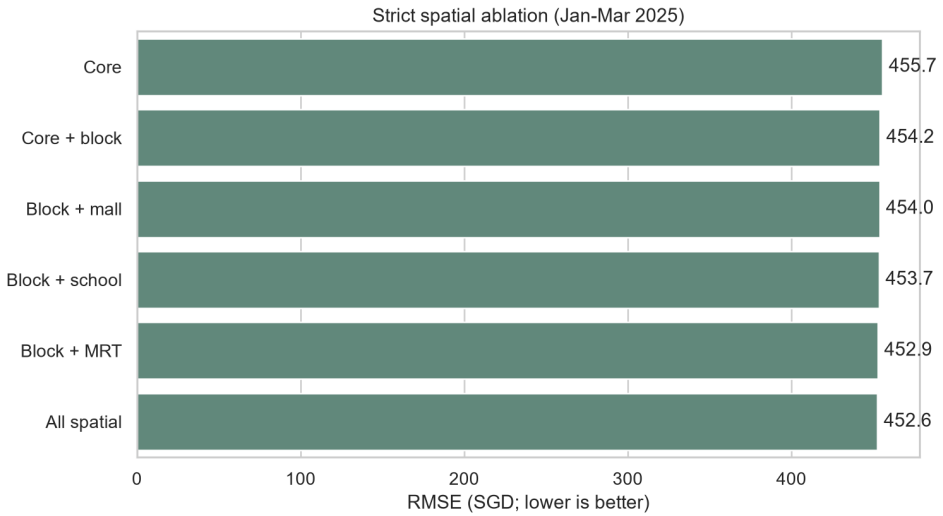
CatBoost leads the initial model screen



All trained models use the same core inputs and rows; the two median rules use raw group summaries. CatBoost uses validation-based early stopping here, so this is an initial screen rather than a fixed-budget family benchmark.

Source: Jan-Mar 2025 validation; Prokhorenkova et al., NeurIPS 2018 for CatBoost

Staged model and feature changes reduced RMSE



Candidate	RMSE
CatBoost core	455.71
Spatial + past rent	450.04
Adaptive market ensemble	445.52
History diversity	440.89
Local trend ensemble	440.76
Recent-window stacking	**440.66**

The left chart isolates spatial additions. The table records successive pipeline versions, so its gains combine feature, target and ensemble changes.

Source: Jan-Mar 2025 validation; later candidates use two seeds and 1,600 trees per component

Current ensemble design

Normal-history view

CatBoost predicts rent directly and as a residual from a past-only market estimate. Local town-flat-type momentum features describe recent changes.

Stale-history view

Eligible training rows receive a second feature view that hides the latest six months of history. The original and augmented copies split that row's weight.

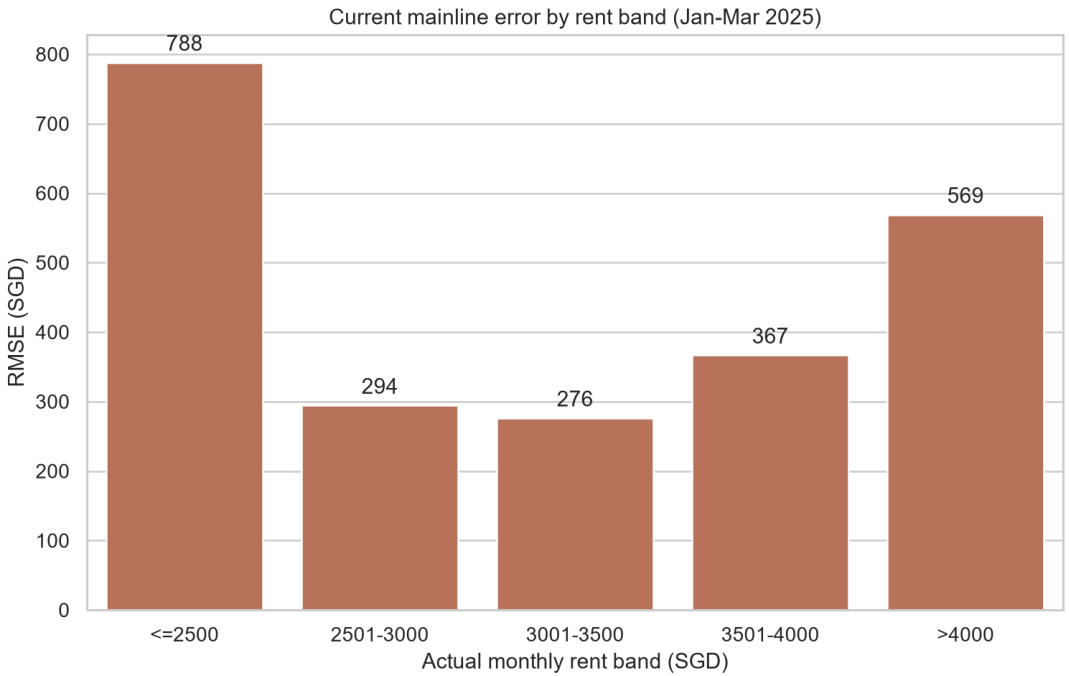
Eight components

Two history views $\times$ two targets $\times$ two random seeds. The mainline averages all components equally. Residual models add the past-only market estimate back at prediction time. Every component uses all permitted history at inference.

Each component uses 1,600 trees, depth 8, learning rate 0.04 and L2 regularisation 6. CatBoost was retained after bounded XGBoost tests and predefined blends failed to improve the same validation windows.

Source: Current configuration, model metadata and frozen stacking study

Current mainline still struggles at rent extremes



- The chart uses the selected 440.76-RMSE mainline on all 8,924 validation rows.
- At or below SGD 2,500, RMSE is 788 with positive bias; this band mixes ordinary low rents and suspected anomalies.
- Rents above SGD 4,000 also remain difficult.
- Jan 2025 RMSE is 456, versus 431 in Mar 2025.

Outlier decision

An earlier test removed 954 suspected low-rent labels and improved the public score by 0.17. The small gain does not confirm that the labels are errors.

Source: Current mainline saved predictions; Jan-Mar 2025 validation; controlled Kaggle submissions

Current results and unresolved limitations

Public submission	RMSE
First submission	438.94488
Adaptive market	436.05684
History diversity	432.47187
Local trend mainline (selected)	432.14881
Stacking candidate*	**432.04286**

*Public best; not promoted.

Total public improvement: **6.90202 RMSE**.

Open risks

- Older-year window: current mainline 590.35; earlier adaptive-market reference 580.54 (9.81 better).
- Weights fitted on Oct–Dec 2024 improve Jan–Mar 2025 by 0.106. The 95% address-bootstrap interval $[-0.034,\ 0.226]$ crosses zero, and transfer is inconsistent.
- Market trend is damped toward six months despite a longer test horizon.
- MRT status is a current snapshot; opening dates are not historical.

Source: Kaggle submission records and temporal validation reports

Next experiments

1. **Market-time extrapolation.** Compare bounded linear, damped and regime-aware forecasts on identical historical cutoffs. Diagnose signed error by forecast horizon.
2. **Nearest comparable rent.** Build a past-only predictor using flat type, similar floor area, nearby blocks and recency. Test it alone before blending.
3. **Independent learner.** Evaluate one bounded LightGBM or alternative model only if its residuals complement CatBoost.
4. **Targeted error analysis.** Decompose gains by town, flat type, rent band, address-history support and month. Avoid rules selected from public leaderboard noise.

Promotion rule

Replace the mainline only when a candidate improves matched recent and long-horizon validation without creating a material regression in an older market regime.

Plan to final submission

Timing	Work and completion criterion
Progress report	Review all claims as a group and submit the PDF.
After feedback	Run the predefined time-extrapolation and nearest-comparable experiments. Record accepted and rejected results.
Model freeze	Select one mainline and one justified alternative using chronological evidence.
Final report	Complete interpretation, references, workload attribution and a clean reproduction.

Current checkpoint

The pipeline is reproducible, leakage-tested and submission-ready. The main unresolved issue is stable prediction across a long, changing market horizon.

AI declaration: We used OpenAI Codex to assist with code implementation and review, experiment execution, slide drafting and language refinement. The project team remains responsible for reviewing the experiments, evidence and conclusions before submission.